Course Offering in FA 2025

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All the courses offered in FA 2025 are categorized into the following three types:

1. **UIUC courses** are those taught by UIUC faculties or co-taught by ZJUI and UIUC faculties. The academic credits (AU) earned from UIUC courses are therefore counted as UIUC credits. Final grades will be recorded in both ZJUI and UIUC systems.
   All UIUC courses are highlighted in blue in the course tables.
2. **ZJUI courses** are those taught by ZJUI faculties normally without the involvement of UIUC faculties. AUs earned from ZJUI courses are therefore counted as ZJUI credits. Grades of these courses will be recorded in INTL PeopleSoft system and will be transferred to UIUC after students have been successfully admitted by UIUC through international transfer during their junior year.
3. **ZJU courses** are those taught by ZJU faculties that are used to fulfil the graduation requirement for the ZJU degree.

Courses are also categorized as mandatory (section 3 in your curriculum) and elective ones (section 4 in your curriculum).

1. **Mandatory courses** include fundamental mathematics and science, technical cores, and composition. Everyone must complete all mandatory courses.
2. **Elective courses** include technical courses, general education (liberal education) and free electives. Elective courses are optional. Students must take sufficient AUs from elective courses to fulfil the total credit requirement of both ZJU and UIUC degrees. Free electives can be almost any course under the categories of ZJUI and UIUC courses.

Please read all the following notes before dive into the course table.

1. All the available courses in FA 2025 are listed below according to the main cohort of attendance.

2. Students are **NOT** allowed to register more than **18 AUs** of the sum of ZJUI and UIUC courses.
3. Students must satisfy the pre-requisites for course registration. The pre-requisites of each course are listed along with course in the following table.
4. **The instructors' name followed by "(Online)" remark, will teach the course online. Except CEE 190 & CEE 201, all the rest of online courses will follow UIUC calendar. First day of instruction will be August 25 2025.**

Courses in FA 2025

1.ZJUI and UIUC courses

A. Freshmen courses

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
ECE 110	Introduction to Electronics	3	Chao QIAN/Umberto Ravaioli (UIUC)	UIUC	
ECE 120	Introduction to Computing	4	Zhang Qiang/Zuozhu LIU/Ujjal Bhowmik (UIUC)	UIUC	
CEE 190	Project Based Introduction to Civil Engineering	4	Art Schmidt (UIUC Online) ZJUI calendar	UIUC	
RHET 101	Principles of Writing	4	Ryan Flanagan/UIUC faculty/ Ashley Barr	UIUC	
CHEM 102	General Chemistry I	3	Fangwei SHAO	ZJUI	
CHEM 103	General Chemistry Lab I	1	Mian PAN/ Miao HUANG	ZJUI	CHEM 102 or concurrent
CS 101	Introduction to Computing: Engineering & Science	3	Weeliat ONG	ZJUI	
ENGR 100	Engineering Orientation	1	Zuozhu LIU/Lee Der-Horng	ZJUI	

			IEE/Erping Li		
MATH 221	Calculus I	4	Cui ZHOU	ZJUI	
MATH 231	Calculus II	3	Cui ZHOU	ZJUI	MATH 221
ME 290	ME Freshmen Seminar	1	Weeliat ONG	ZJUI	
PHYS 211	University Physics: Statics	4	Oleksiy Penkov	ZJUI	

1. **AU cap**: Semester AU cap in Fall 2025 is set at 20 AU for all the freshmen.
2. **TPT for MATH 221**: Students passing Math 221 proficiency exam will proceed to Math 231 and can also register to MATH 257 if AU cap allows.
3. **CivE required**: ENGR 100, Math 221, CEE 190, RHET 101, PHYS 211. Student can register CS 101 or CHEM 102, if s/he has not reached AU cap.
4. **CompE required**: ENGR 100, Math 221, ECE 110, ECE 120, RHET 101. Student can register CHEM 102/103, if s/he has not reached AU cap.
5. **EE required**: ENGR 100, CHEM 102/103, Math 221, RHET 101, ECE 110. Student can register ECE 120, if s/he has not reached AU cap.
6. **ME required**: ENGR 100, Math 221, CS 101, ME 290, RHET 101, PHYS 211. Student can register CHEM 102/103, if s/he has not reached AU cap.
7. CS 101 and CHEM 102/103 are recommended for CompE and EE students as free elective courses.
8. ECE 110, ECE 120, CS 101, CHEM 102/103 and Phys 211 will be offered again in SP 2026.
9. All required courses are pre-registered. Student cannot drop these courses unless approved by institute.

B. Sophomore Courses

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
CEE 201	Systems Engineering & Economics	3	Hadi Meidani (UIUC Online) ZJUI calendar	UIUC	MATH231&CS101& or concurrent
ME 270	Design for Manufacturability	3	Timothy LEE/Mickey Clemon	UIUC	ME170

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
TAM 211	Statics	3	Jiahuan CUI/Tom Golecki	UIUC	PHYS211&((MATH concurrent) or (MA concurrent))
ECE 313	Probability with Engineering Application	3	Piao CHEN/Xu Chen	UIUC	MATH257 or concu
ECE 314	Probability in Engineering Lab	1	Piao CHEN/Xu Chen	UIUC	ECE313 or concurre
MATH 257	Linear Algebra with Computational Applications	3	Xiangming MENG	ZJUI	MATH221 & (CS10 120 or concurrent)
MATH 213	Introduction to Discrete Mathematics	3	Meng ZHANG	ZJUI	MATH221
MATH 241	Calculus III	4	Hao YANG/Qingchun HOU	ZJUI	MATH231
PHYS 212	University Physics: Elec& Mag	4	Ruisheng DIAO/ Chao QIAN	ZJUI	PHYS211

1. **AU cap**: Semester AU cap in Fall 2025 is set at 18 AU for all the sophomore.
2. **CivE required**: MATH 257, MATH 241, PHYS 212, CEE 201, TAM 211.
3. **ME required**: MATH 257, MATH 241, PHYS 212, ME 270, TAM 211.
4. **CompE required**: MATH 213, MATH 257, MATH 241, PHYS 212. ECE 313/314 are highly recommended.
5. **EE required**: MATH 257, MATH 241, PHYS 212. ECE 313/314 are highly recommended. Math 213 is highly recommended, but not required.
6. Sophomore students can enroll other courses if AU cap is not exceeded.

C. Junior Courses

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
ECE329	Fields and Waves I	3	Shurun TAN	ZJUI	ECE210

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
ECE374	Introduction to Algorithms & Models of Computation	4	Pavel LOSKOT	ZJUI	(CS 173 or MATH 213)& CS 225

1. **AU cap**: Semester AU cap in Fall 2025 is set at 18 AU for all the juniors.
2. All the junior students staying on Haining campus in FA 2025 should attend academic advisory interview at UA office. Please make an appointment via zjui_undergraded@intl.zju.edu.cn

D. Senior Courses

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
ECE 408	Applied Parallel Programming	4	Volodymyr Kindratenko (UIUC Online) UIUC calendar	UIUC	ECE 220
ECE 457	Microwave Devices & Circuits	3	Hanzhi MA	ZJUI	ECE 329 or concurrent, & ECE 350 is recommended
ECE 459	Communications System	3	Said MIKKI/Juan Alvarez (UIUC)	UIUC	ECE 313
ECE 464	Power Electronics	3	Chushan LI/Arijit Banerjee (UIUC)	UIUC	
ECE 469	Power Electronics Lab	2	Lin QIU/Arijit Banerjee (UIUC)	UIUC	ECE 343 & (ECE 464 or concurrent)
ECE 470/ ME 445*	Introduction to Robotics	4	Liangjing YANG/Katie Driggs-Campbell (UIUC)	UIUC	MATH 225 or MATH 257 or MATH 418
ECE 486	Control Systems	4	Hua CHEN/Ali Belabbas (UIUC)	UIUC	ECE210 or ME 360
ECE 490	Introduction to Optimization	3	Bo ZHAO	ZJUI	ECE 220 & MATH

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
					257
ECE 498MF	Mathematical Foundations for Machine Learning	3	Pavel Loskot	ZJUI	Math 213 & Math 257 & (Math 285 or 286)
ECE 498EAI	Embodied Artificial Intelligence	3	Gaoang WANG	ZJUI	Math 257 & ECE 313
CS 442	Trustworthy machine learning	3	Zuozhu LIU/Yushi CHENG	ZJUI	CS225 & (Math221 or Math257)
CS 450	Numerical Analysis	3	Hao YANG	ZJUI	(CS101 or ECE 120 or ECE 220) & MATH 285 & (MATH257 or MATH225)
ME 320	Heat Transfer	4	Weeliat ONG/Sanjiv Sinha (UIUC)	UIUC	ME200 & (ME310 or TAM335) & MATH285
ME 371	Mechanical Design II	3	Timothy LEE/Brian Mercer (UIUC)	UIUC	(CEE330 or (ME330 or concurrent)) & ME370
ME 412	Numerical Thermo-Fluid Mechanics	3	Jiahuan CUI	ZJUI	(ME310 or TAM335) & (ME320 or concurrent)
ME 453	Data Science in Manufacturing Quality Control	3	R. Toro Santamaria (UIUC online) UIUC calendar	UIUC	ME270 & (STAT400 or IE300) & MATH257
~~ME 471~~	~~Finite Element Analysis~~	~~3~~	~~Yasutaka Narazaki~~	~~ZJUI~~	~~CS 101 & ME 371 or TAM 470~~
CEE 416	Traffic Capacity Analysis	3	Simon HU/Ray Benekohal (UIUC)	UIUC	CEE310

COURSE CODE	COURSE NAME	AU	INSTRUCTORS	COURSE TYPE	Pre-requisite
CEE 434	Environmental Systems I	3	Tingju ZHU/Ximing Cai (UIUC)	UIUC	CEE350
CEE 460	Steel Structures I	3	Zhi LI	ZJUI	CEE360
CEE 470	Structural Analysis	4	Binbin LI/Bill Spencer (UIUC)	UIUC	CEE360
CEE 495	Professional Practice	0	Jeff Roesler (UIUC Online) UIUC calendar	UIUC	

1. **CivE required**: CEE 495, CEE 460 and CEE 470 (structure area), CEE 434 (water resource area), CEE 416 (transportation area).
2. **ME required**: ME 320, ME 371.
3. CS 450 can be used by CivEs as a Sci Elective. CS 450 will be transferred to UIUC as CS 357. Students cannot receive credits for both CS 450 and CS 357.
4. Senior students should check degree audit in PS system and enroll the technical electives/GedEd courses according to the curriculum requirements of your major.
5. Students that enroll ECE 464, are required to enroll ECE 469 concurrently.
6. CS 442 can be used as a free elective course.
7. CS 3-- and 4-- courses, such as CS 412, CS 442 and CS 450, cannot be used to confer CS minor, if it is a transferred credit.
8. For ECE 470/ME 445, EE and CompE students should enroll ECE 470, if choose this course. ME student should enroll ME 445.
9. ECE 498MF, ECE 498EAI and CS 442 still need further confirmation from UIUC before they are used as tech electives.

E. Liberal Education courses (GenEd)

1. Liberal Education courses fall into various categories.

   HUM - Humanities and the Arts

   a. **HP** - Historical and Philosophical Perspectives

   b. **LA** - Literature and the Arts

   SBS - Social and Behavioral Sciences

a. **BSC** - Behavioral Sciences

b. **SS** - Social Sciences

CS - Cultural Studies

a. **NW** - Non-Western Culture(s)

b. **US** - US Minority Cultures

c. **WCC** - Western/Comparative Culture(s)

2. You need to enroll the GenEd courses that satisfy both total AU number and AUs in each category and subcategory. For the GenEd you are required to satisfy, please check “Liberal Education” in section 4 of your curriculum.
3. You can take GenEd courses in any semester at either ZJUI or UIUC, as long as the semester AU cap permits.
4. All the GenEd course offered on Haining campus are ZJUI courses and AU is counted into total AU number for semester cap.
5. One (1) Aesthetic Education (AE) courses are required by ZJU degree.
6. The GenEd courses offered in FA 2025 is shown as following:

No.	Course Number	Course Titles	Credits	Categories	Language	Name
1	CLCV 115*	Mythology of Greece and Rome	3	WCC/LA/AE	English	朱振宇 Zhu Zhengyu
2	ECON 102	Introduction to Microeconomics	3	SS	English	周默涵 Zhou Mohan
3	PHIL 101	Introduction to Philosophy	3	HP	English	曾劭愷 Tseng Shaokai
4	ECON 103	Macroeconomic Principles	3	SS	English	叶兵 Ye Bing
5	CMN 280	Communication, Technology and Society	3	SS	English	赵瑜佩 Zhao Yupei
6	UNWL 316*	Modern Chinese Literature	3	NW/LA/AE	Chinese	张广海 Zhang

						Guanghai

- *Count as ZJU Aesthetic Education course.

2. ZJU courses

FALL Semester	Chinese/ International students	Course Code	Course Title
Freshmen	Chinese students	MITR1001	军训/Military Training
Freshmen	Chinese students	PE1001	体育Ⅰ/Physical Education I
Freshmen	Chinese students	PS1001A	形势与政策Ⅰ/Situation and Policy I
Freshmen	Chinese students	ENGL1001	综合英语Ⅰ/Integrated English I
Freshmen	Chinese students	PHIL2002	马克思主义基本原理/Introduction to the Principle of Marxism
Freshmen	Chinese students	PS2011	毛泽东思想和中国特色社会主义理论体系概论/ Intro.to Mao Thought & Theoretical System of China Socialism
Freshmen	International students	CHIN1001	基础汉语Ⅰ/Chinese Ⅰ
sophomore	Chinese students	PS2011	毛泽东思想和中国特色社会主义理论体系概论/ Intro.to Mao Thought & Theoretical System of China Socialism
sophomore	Chinese students	PE2001	体育Ⅲ/Physical Education Ⅲ
sophomore	Chinese students	PS2001A	形势与政策Ⅱ/Situation and Policy Ⅱ
sophomore	Chinese students	MITR2001	军事理论/Military Theory
sophomore	International students	CHIN1003	基础汉语Ⅲ/Chinese Ⅲ
sophomore	International students	CULT2001	中国概况
Junior	Chinese students	PS2001C	形势与政策Ⅱ/Situation and Policy Ⅱ

senior	Chinese students	PS3011	习近平新时代中国特色社会主义思想概论/General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era
senior	Chinese students	PE4021	体育Ⅶ/Physical Education Ⅶ--Fitness test and exercise
senior	Chinese students	PS2001E	形势与政策 II /Situation and Policy II

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